

Lecture 15 - ANCOVA

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Lecture 14: Review

Review

General Linearized Models (GLM)

- Gaussian GLMs (normal distribution)
- Poisson GLMs (count data)
- Logistic GLMs (binary outcomes)
- Model assumptions and diagnostics
- Connection to ANOVA and linear models

Lecture 15: ANCOVA Overview

Overview

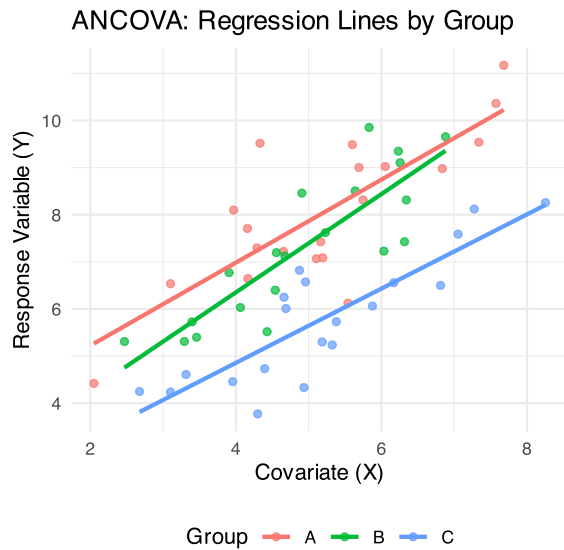
Analysis of covariance (ANCOVA):

- Introduction to ANCOVA
- When to use ANCOVA
- Linear model for ANCOVA
- Analysis of variance in ANCOVA
- Assumptions of ANCOVA
 - Homogeneous slopes
- Robust ANCOVA approaches
- Specific comparisons of means
- Examples and interpretation
- Scientific reporting of ANCOVA results

Introduction to ANCOVA

What is ANCOVA?

- ANCOVA = Analysis of COVariance
- Combination of regression and ANOVA
- A continuous covariate is measured along with the response variable for each experimental unit
- Common use: compare means of factor levels (groups), adjusting for variance from continuous covariate
- Another use: determine whether two or more regression lines differ in slopes and intercepts



When to Use ANCOVA

Common Applications of ANCOVA

- **Increasing statistical power**
 - Removing variation associated with a covariate can reduce residual error
 - More powerful test of treatment effects
- **Adjusting for confounding variables**
 - When treatments might differ in some continuous variable
 - Need to separate treatment effects from covariate effects
- **Testing equality of regression lines**
 - Do treatments have the same relationship with a continuous variable?
 - Tests for both slopes and intercepts

ANCOVA Example: Cricket Chirping

Cricket Chirping Example

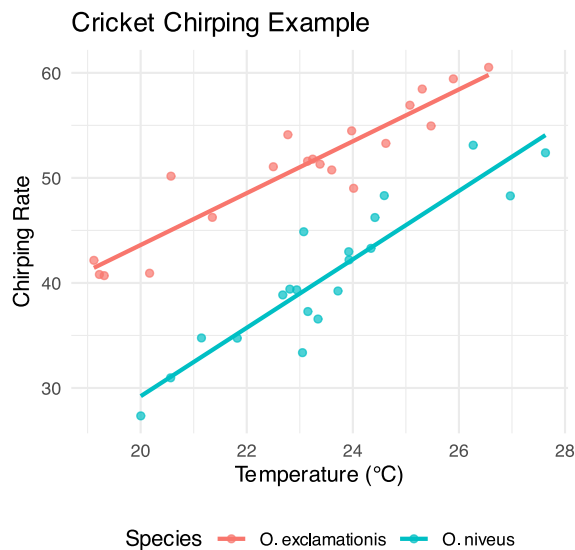
Want to compare chirping rate of two cricket species:

- *Oecanthus exclamationis*
- *Oecanthus niveus*

But:

- Measured rates at different temperatures
- Range of temperatures differed between species
- Apparent relationship between pulse rate and temperature

ANCOVA lets us adjust for temperature effect to get a more powerful test!



ANCOVA Linear Model: Conceptual Framework

The ANCOVA Model

Key concept in ANCOVA: the difference between “unadjusted” group means and “adjusted” means.

In this visualization:

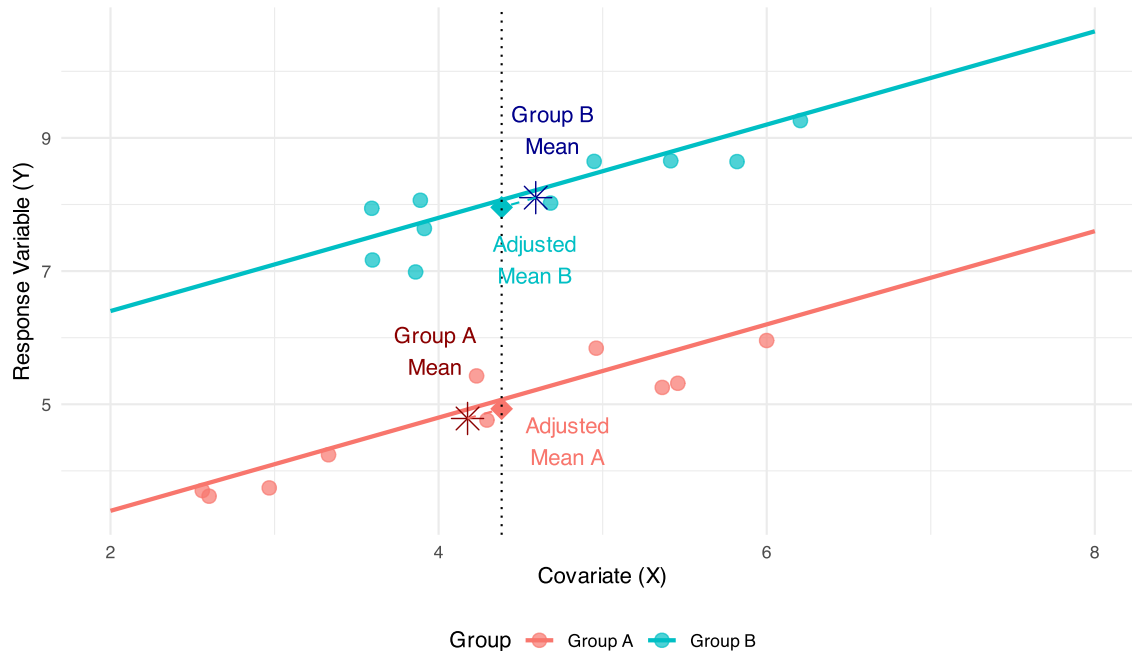
- Group Means (shown as asterisks): raw/unadjusted means for each group - simply the average X value and average Y value for all points in that group. Notice that Group A and Group B have different mean X values (they’re positioned at different points along the X axis).
- Adjusted Means (shown as triangles): These are what ANCOVA actually compares. The adjusted means represent what each group’s mean would be if all groups had the same value of the covariate (in this case, the overall mean X).

The core purpose of ANCOVA is to make this adjustment. This is important because:

- When groups differ in their covariate values (as they often do in observational studies or even in experiments with random assignment), comparing raw means can be misleading
- The adjustment helps “level the playing field” by estimating what each group’s mean would be if they all had the same value of the covariate

ANCOVA Model Visualization

ANCOVA: Adjusting Means Based on Covariate



Mathematical Model for ANCOVA

For a single-factor ANCOVA with factor A (p levels, $i = 1$ to p), a continuous covariate (x), and response variable (y):

$$Y_{ij} = \mu + \alpha_i + \beta(X_{ij} - \bar{X}) + \varepsilon_{ij}$$

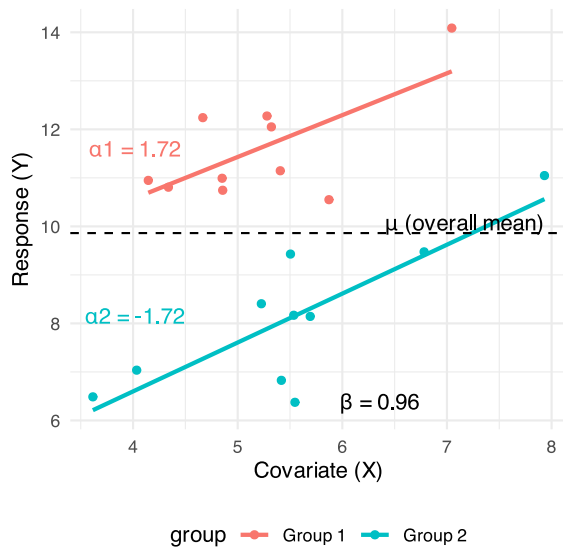
Where: - Y_{ij} = response value for observation j in level i of factor A - μ = overall mean - α_i = effect of level i of factor A - β = common regression slope relating Y to X - X_{ij} = covariate value for observation j in level i of factor A - $\bar{X}$ = mean value of covariate - ε_{ij} = error term

ANCOVA Parameters Interpretation

Interpretation of Parameters

- μ = overall mean response
- α_i = effect of level i (difference between group mean and overall mean)
- β = pooled within-group regression coefficient
- X_{ij} = covariate value for observation j in group i
- $\bar{X}$ = overall mean of covariate
- ε_{ij} = unexplained error

This model assumes **homogeneous slopes** across all treatment groups (we'll test this later).



ANCOVA in R: Basic Implementation

Running ANCOVA in R

Basic ANCOVA model: - Response: continuous variable (y) - Predictor: categorical factor (A)
 - Covariate: continuous variable (x)

The simplest ANCOVA model is:

```
model <- lm(y ~ A + x, data = mydata)
anova(model)
```

Alternative: use aov() function

```
model <- aov(y ~ A + x, data = mydata)
summary(model)
```

Both approaches use Type I SS (sequential). For unbalanced designs, you may want Type III SS using car package.

```
# Load the partridge dataset
partridge <- read.csv("data/partridge.csv")

# Look at the first few rows
head(partridge)
```

	PARTNERS	TYPE	TREATMEN	LONGEV	LLONGEV	THORAX	RESID1	PREDICT1	RESID2
1	8	0	1	35	1.544068	0.64	-5.868456	40.86846	-0.04743024
2	8	0	1	37	1.568202	0.68	-9.301196	46.30120	-0.07105067
3	8	0	1	49	1.690196	0.68	2.698804	46.30120	0.05094369
4	8	0	1	46	1.662758	0.72	-5.733936	51.73394	-0.02424867
5	8	0	1	63	1.799341	0.72	11.266064	51.73394	0.11233405
6	8	0	1	39	1.591065	0.76	-18.166676	57.16668	-0.14369601
PREDICT2									
1	1.591498								
2	1.639252								
3	1.639252								
4	1.687007								

```
5 1.687007
6 1.734761
```

```
# Create factors for treatment
partridge$TREATMEN <- as.factor(partridge$TREATMEN)

# Basic ANCOVA model
model1 <- lm(LONGEV ~ THORAX + TREATMEN,
             data = partridge)

# View ANOVA table with Type I SS
anova(model1)
```

Analysis of Variance Table

```
Response: LONGEV
      Df Sum Sq Mean Sq F value    Pr(>F)
THORAX   1 15496.6  15496.6 140.293 < 2.2e-16 ***
TREATMEN  4   9611.5   2402.9   21.753 1.719e-13 ***
Residuals 119 13144.7    110.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANCOVA in R: Type III Sum of Squares

Using Type III SS from car package:

```
# Using Type III SS from car package
model2 <- lm(LONGEV ~ TREATMEN + THORAX,
             data = partridge)
Anova(model2, type = "III")
```

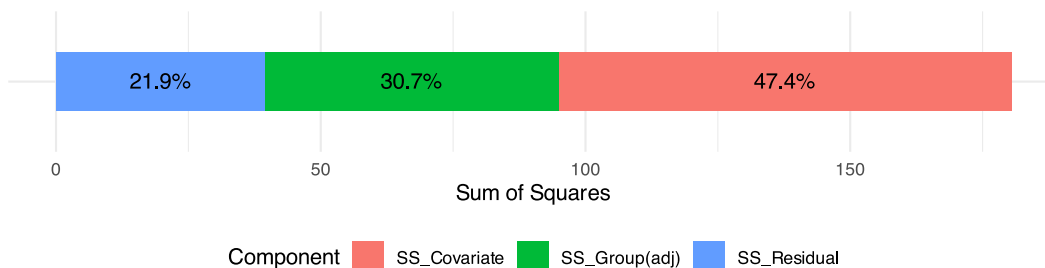
Anova Table (Type III tests)

```
Response: LONGEV
      Sum Sq Df F value    Pr(>F)
(Intercept) 2234.9   1  20.233 1.605e-05 ***
TREATMEN     9611.5   4  21.753 1.719e-13 ***
THORAX      13168.9   1 119.219 < 2.2e-16 ***
Residuals   13144.7 119
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

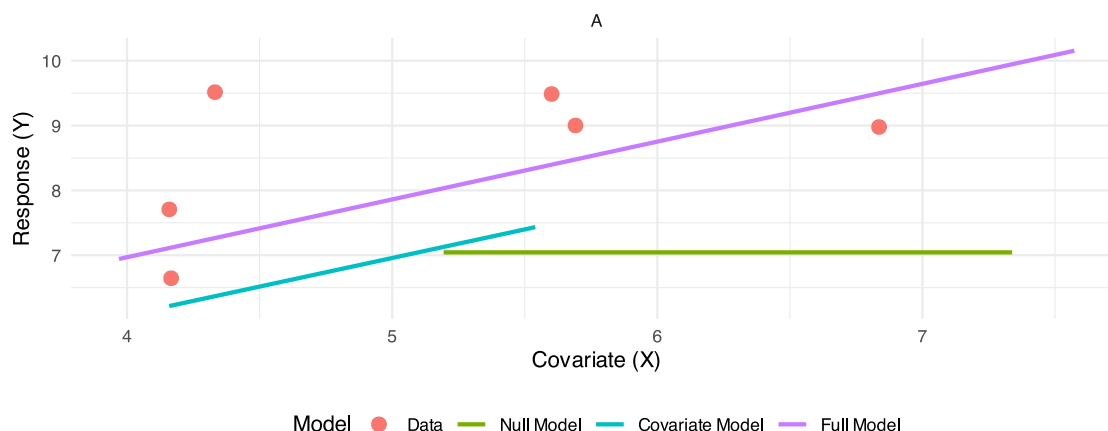
The Type III approach is often preferred for: - Unbalanced designs - When you want to test each effect adjusted for all others - More conservative approach when groups differ in covariate values

Analysis of Variance for ANCOVA: Partitioning

Partitioning of Variance in ANCOVA



Model Fits in ANCOVA



ANOVA Table for ANCOVA

The ANOVA table for a single-factor ANCOVA has these components:

Source	df	Sum of Squares	Mean Square	F-ratio	Expected MS
Factor A (adjusted)	(p-1)	SS_A(adj)	$MS_A(adj) = SS_A(adj)/(p-1)$	$MS_A(adj)/MS_{Residual}$	$\sigma^2 + n \sum \alpha^2 / (p-1)$
Covariate	1	SS_Covariate	$MS_{Covariate} = SS_{Covariate}/1$	$MS_{Covariate}/MS_{Residual}$	$\sigma^2 + \beta^2 \sum (X-X)^2$
Residual	n-p-1	SS_Residual	$MS_{Residual} = SS_{Residual}/(n-p-1)$		σ^2
Total	n-1	SS_Total			

Null Hypotheses in ANCOVA

1. **Treatment Effect (adjusted for covariate)** $H_0 : \alpha_1 = \alpha_2 = \dots = \alpha_p = 0$

- Are the adjusted group means equal?
- Test with $F = MS_A(adj)/MS_{Residual}$

2. **Covariate Effect** $H_0 : \beta = 0$

- Is there a relationship between the covariate and the response?
- Test with $F = MS_{Covariate}/MS_{Residual}$

3. **Homogeneity of Slopes** (test this first!) $H_0 : \beta_1 = \beta_2 = \dots = \beta_p$

- Are the regression slopes the same for all groups?
- Test by adding group*covariate interaction term

Testing Homogeneity of Slopes

Testing for Homogeneous Slopes

ANCOVA assumes the regression slopes are the same for all groups (parallel regression lines)

To test this assumption:

1. Fit model with interaction term:

```
model_int <- lm(y ~ A * x, data = mydata)
```

2. Test significance of interaction:

```
anova(model_int)
```

3. If interaction is significant ($p < 0.05$):

- Slopes are not homogeneous
- Standard ANCOVA inappropriate
- Use alternative approaches

```
# Test homogeneity of slopes in partridge data
# by adding interaction term
model_int <- lm(LONGEV ~ TREATMEN * THORAX,
               data = partridge)

# Test significance of interaction
anova(model_int)
```

Analysis of Variance Table

Response: LONGEV

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
TREATMEN	4	11939.3	2984.8	26.1983	1.896e-15	***
THORAX	1	13168.9	13168.9	115.5855	< 2.2e-16	***
TREATMEN:THORAX	4	42.5	10.6	0.0933	0.9844	
Residuals	115	13102.1	113.9			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Extract p-value for interaction
interaction_pvalue <- anova(model_int)[3, "Pr(>F)"]
```

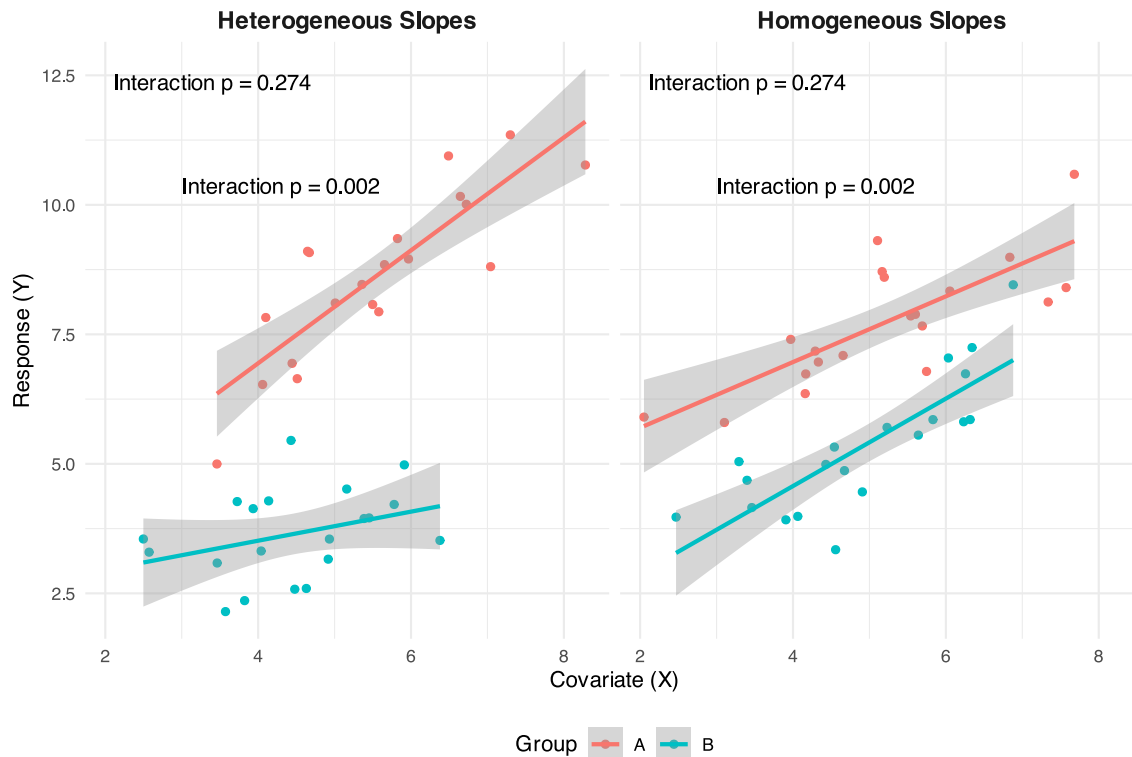
The p-value for the interaction is 0.984. Since $p > 0.05$, we can proceed with standard ANCOVA (assuming homogeneous slopes).

Visualization of Homogeneity of Slopes

Parallel vs. Non-Parallel Slopes

Testing Homogeneity of Slopes in ANCOVA

Left: Parallel slopes (homogeneous) | Right: Non-parallel slopes (heterogeneous)



Handling Heterogeneous Slopes

When Slopes Are Not Homogeneous

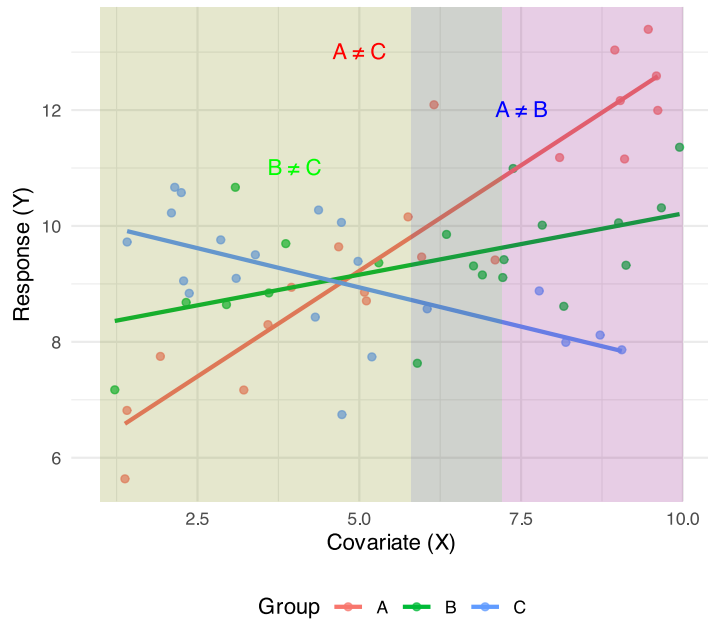
If the interaction term is significant ($p < 0.05$), the slope-group relationship is not the same across groups.

Options:

1. **Report the interaction** - this is a biologically interesting result!
2. **Separate regressions** - analyze each group separately
3. **Johnson-Neyman procedure** - identifies regions of the covariate where groups differ significantly
4. **Alternative models** - consider transformation, polynomial terms, or more complex models

Heterogeneous Slopes Example

Johnson-Neyman Regions of Significance



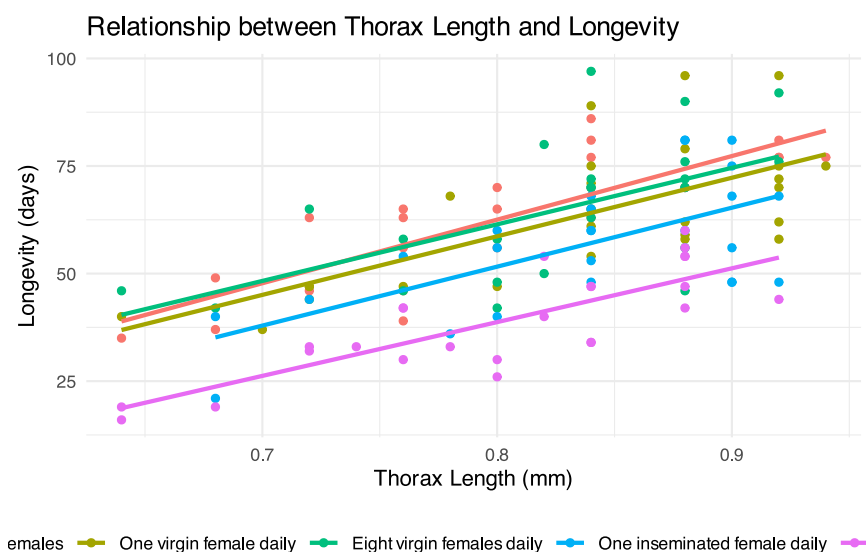
Partridge Example: Data Overview

ANCOVA on Longevity of Male Fruitflies

```
# 1. Examine the partridge data structure
# str(partridge)

# Create better names for treatments
partridge$treatment <- factor(partridge$TREATMEN,
                              levels = 1:5,
                              labels = c("No females",
                                          "One virgin female daily",
                                          "Eight virgin females daily",
                                          "One inseminated female daily",
                                          "Eight inseminated females daily"))

# 2. Create a plot of the data showing relationship
ggplot(partridge, aes(x = THORAX, y = LONGEV, color = treatment)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Relationship between Thorax Length and Longevity",
       x = "Thorax Length (mm)",
       y = "Longevity (days)",
       color = "Treatment") +
  theme_minimal() +
  theme(legend.position = "bottom")
```



Partridge Example: Testing Homogeneity

Testing Homogeneity of Slopes

```
# Test for homogeneity of slopes
homo_slopes_model <- lm(LONGEV ~ THORAX * treatment, data = partridge)
anova(homo_slopes_model)
```

Analysis of Variance Table

Response: LONGEV

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
THORAX	1	15496.6	15496.6	136.0170	< 2.2e-16 ***
treatment	4	9611.5	2402.9	21.0905	4.617e-13 ***
THORAX:treatment	4	42.5	10.6	0.0933	0.9844
Residuals	115	13102.1	113.9		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
# Extract the p-value for the interaction
p_interaction <- anova(homo_slopes_model)[3, "Pr(>F)"]
```

The p-value for the interaction term (treatment × THORAX) is 0.984. Since this value is > 0.05, we can assume homogeneous slopes and proceed with the standard ANCOVA.

Partridge Example: Full ANCOVA Analysis

```
# Fit the ANCOVA model (without interaction)
ancova_model <- lm(LONGEV ~ THORAX + treatment, data = partridge)

# View ANOVA table
anova(ancova_model)
```

Analysis of Variance Table

```
Response: LONGEV
      Df Sum Sq Mean Sq F value    Pr(>F)
THORAX   1 15496.6 15496.6 140.293 < 2.2e-16 ***
treatment 4  9611.5  2402.9  21.753 1.719e-13 ***
Residuals 119 13144.7   110.5
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Get more detailed summary
summary(ancova_model)
```

```
Call:
lm(formula = LONGEV ~ THORAX + treatment, data = partridge)

Residuals:
    Min       1Q   Median       3Q      Max
-26.189  -6.599  -0.989   6.408  30.244

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    -46.055     10.239  -4.498 1.61e-05
THORAX          135.819     12.439  10.919 < 2e-16
treatmentOne virgin female daily    -3.929      2.997  -1.311 0.192347
treatmentEight virgin females daily  -1.276      2.983  -0.428 0.669517
treatmentOne inseminated female daily -10.946      2.999  -3.650 0.000391
treatmentEight inseminated females daily -23.879      2.973  -8.031 7.83e-13

(Intercept)          ***
THORAX                ***
treatmentOne virgin female daily
treatmentEight virgin females daily
treatmentOne inseminated female daily ***
treatmentEight inseminated females daily ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.51 on 119 degrees of freedom
Multiple R-squared:  0.6564,    Adjusted R-squared:  0.6419
F-statistic: 45.46 on 5 and 119 DF,  p-value: < 2.2e-16
```

```
# Get adjusted means using emmeans
adjusted_means <- emmeans(ancova_model, "treatment")
adjusted_means
```

treatment	emmean	SE	df	lower.CL	upper.CL
No females	65.4	2.11	119	61.3	69.6
One virgin female daily	61.5	2.11	119	57.3	65.7
Eight virgin females daily	64.2	2.10	119	60.0	68.3
One inseminated female daily	54.5	2.11	119	50.3	58.7
Eight inseminated females daily	41.6	2.12	119	37.4	45.8

```
Confidence level used: 0.95
```

Partridge Example: Pairwise Comparisons

Pairwise Comparisons of Adjusted Means

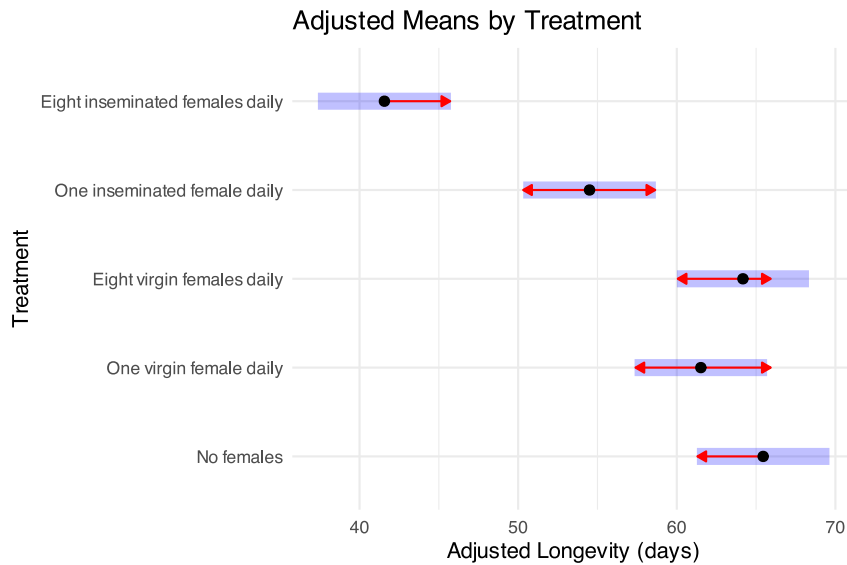
```
# Pairwise comparisons of adjusted means  
pairs(adjusted_means, adjust = "tukey")
```

contrast	estimate	SE
No females - One virgin female daily	3.93	3.00
No females - Eight virgin females daily	1.28	2.98
No females - One inseminated female daily	10.95	3.00
No females - Eight inseminated females daily	23.88	2.97
One virgin female daily - Eight virgin females daily	-2.65	2.98
One virgin female daily - One inseminated female daily	7.02	2.97
One virgin female daily - Eight inseminated females daily	19.95	3.01
Eight virgin females daily - One inseminated female daily	9.67	2.98
Eight virgin females daily - Eight inseminated females daily	22.60	2.99
One inseminated female daily - Eight inseminated females daily	12.93	3.01

df	t.ratio	p.value
119	1.311	0.6849
119	0.428	0.9929
119	3.650	0.0035
119	8.031	<.0001
119	-0.891	0.8996
119	2.361	0.1336
119	6.636	<.0001
119	3.249	0.0129
119	7.560	<.0001
119	4.298	0.0003

P value adjustment: tukey method for comparing a family of 5 estimates

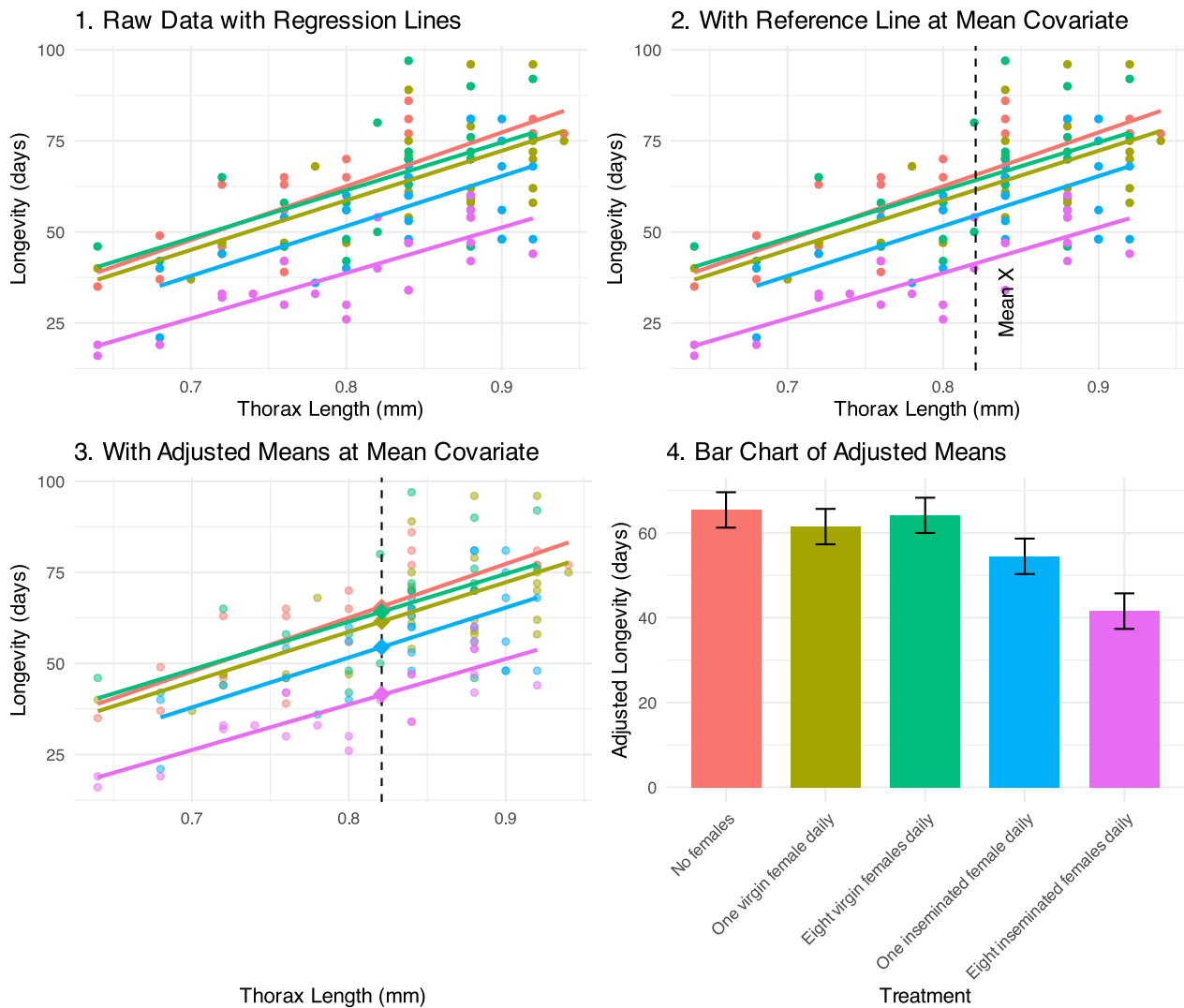
```
# Plot adjusted means with confidence intervals  
plot(adjusted_means, comparisons = TRUE) +  
  labs(title = "Adjusted Means by Treatment",  
        x = "Adjusted Longevity (days)",  
        y = "Treatment") +  
  theme_minimal()
```



Visualizing ANCOVA Results

Visualization Options for ANCOVA

Visualization Options for ANCOVA Results

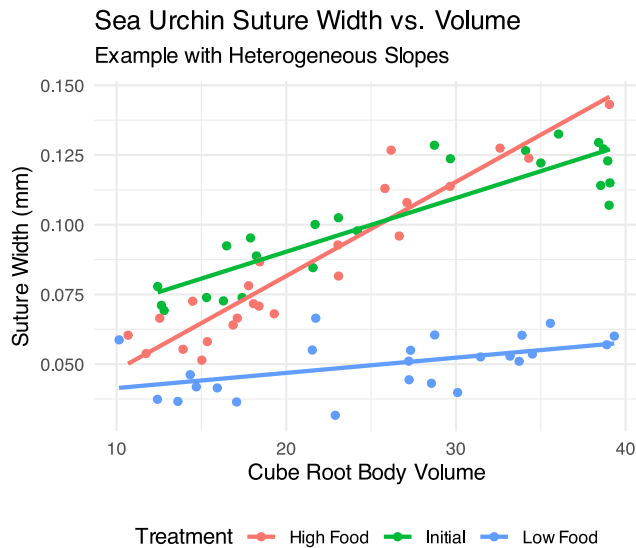


Sea Urchin Example: Heterogeneous Slopes

Heterogeneous Slopes Example

Constable (1993) studied shrinking in sea urchin test: - Compared suture widths between treatments - Three groups: high food, low food, initial sample - Covariate: body volume (cube root transformed)

The analysis showed: - Significant interaction between treatment and covariate - **Heterogeneous slopes** across treatments - Can't use standard ANCOVA



Johnson-Neyman Procedure

Johnson-Neyman Procedure for Heterogeneous Slopes

Analysis of Variance Table

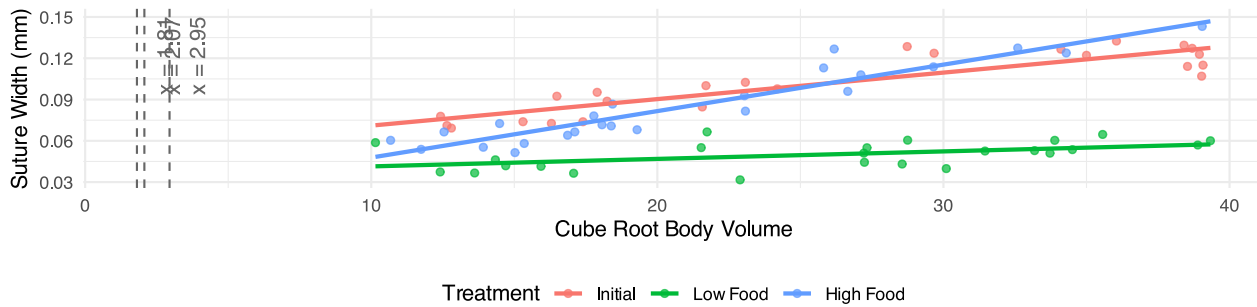
Response: suture_width

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
volume	1	0.015724	0.0157238	176.91	< 2.2e-16 ***
treatment	2	0.036482	0.0182411	205.23	< 2.2e-16 ***
volume:treatment	2	0.006213	0.0031064	34.95	4.453e-11 ***
Residuals	66	0.005866	0.0000889		

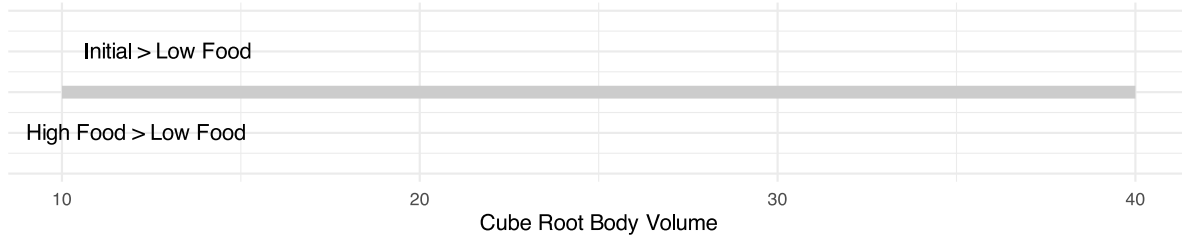
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Johnson-Neyman Procedure for Heterogeneous Slopes

Identifies regions where treatments differ significantly



Regions of Significant Differences



Interpretation of Heterogeneous Slopes

When you have heterogeneous slopes, the Johnson-Neyman procedure identifies regions of the covariate where groups differ:

1. **Initial > Low Food** when cube root body volume > 2.95
 - For large urchins, the initial sample has wider sutures than low food urchins
2. **High Food > Initial** when cube root body volume > 1.81
 - For most urchins, high food treatment results in wider sutures than initial samples
3. **High Food > Low Food** when cube root body volume > 2.07
 - For most medium to large urchins, high food results in wider sutures than low food

The biological interpretation is that **food regime affects suture width differently depending on urchin size**. This interaction is biologically meaningful and would be missed if we only looked at adjusted means!

Assumptions of ANCOVA

Key Assumptions

1. **Independence of observations**
 - Samples are random and independent
 - No clustering or repeated measures
2. **Normal distribution of residuals**
 - Residuals follow a normal distribution
 - Check with QQ plots or formal tests
3. **Homogeneity of variances**
 - Equal variances across groups
 - Check with residual plots vs. fitted values
4. **Linearity**
 - Linear relationship between Y and X within each group
 - Check with scatterplots
5. **Homogeneity of regression slopes** (critical!)
 - Regression slopes equal across all groups

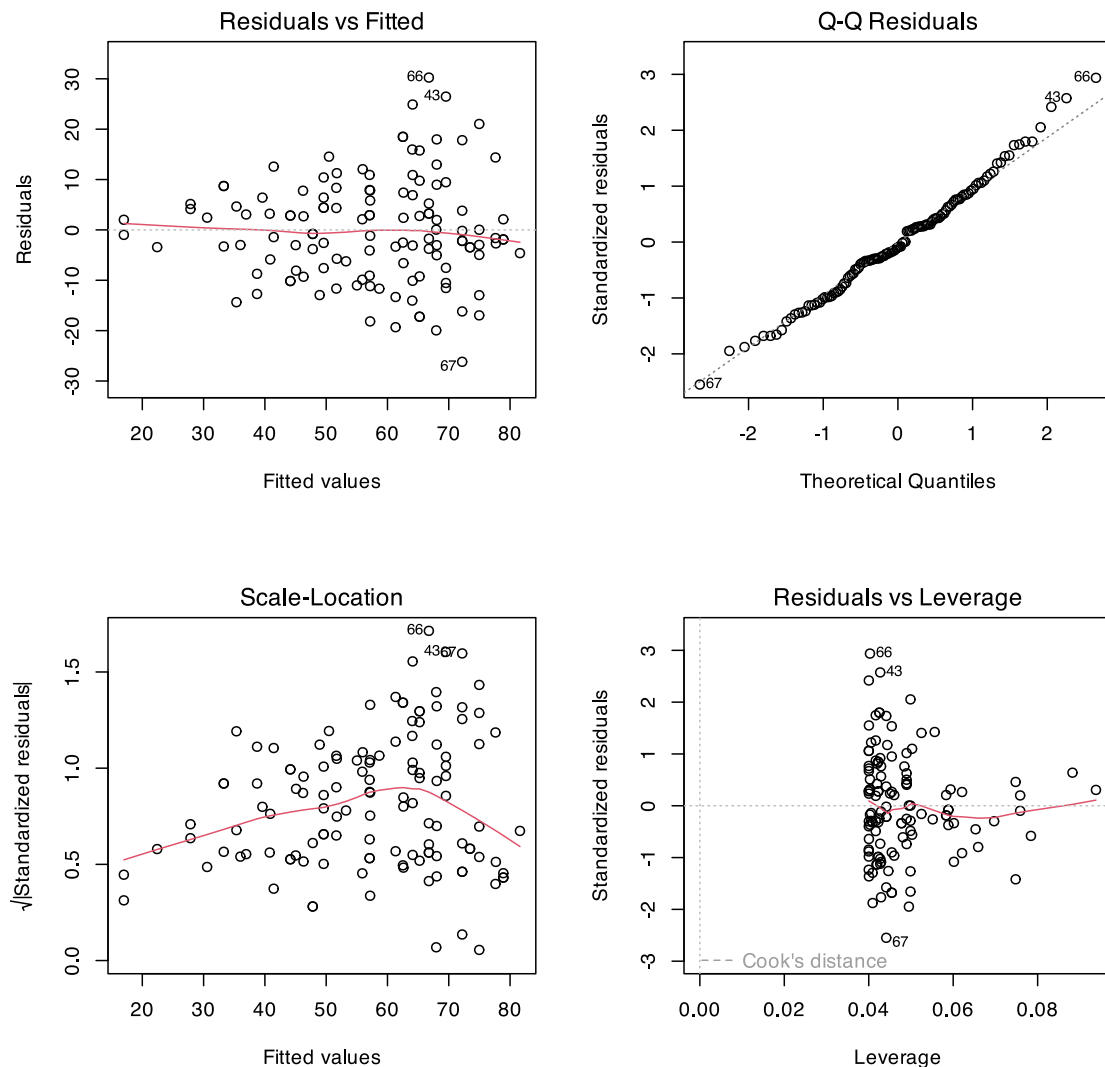
- Test with interaction term

Checking Assumptions in R

Checking Assumptions in R

```
# Fit ANCOVA model for partridge data
ancova_model <- lm(LONGEV ~ THORAX + treatment, data = partridge)

# Create a 2x2 panel of diagnostic plots
par(mfrow = c(2, 2))
plot(ancova_model)
```



Robust ANCOVA Approaches

Non-Parametric Alternatives

When ANCOVA assumptions are violated, consider:

1. Rank Transformation

- Rank transform both Y and X variables
- Run standard ANCOVA on ranked data

- Simple but may not handle interactions well
2. **ANCOVA on Bootstrapped Data**
 - Use bootstrapping to estimate parameters
 - Doesn't require normality assumption
 3. **Quantile Regression**
 - Models relationships at different quantiles
 - Robust to outliers and heteroscedasticity
 4. **Permutation Tests**
 - Randomization tests of treatment effects
 - No distributional assumptions

```
# Example of rank-transformed ANCOVA
partridge$rank_LONGEV <- rank(partridge$LONGEV)
partridge$rank_THORAX <- rank(partridge$THORAX)

# Fit ANCOVA on ranked data
rank_ancova <- lm(rank_LONGEV ~
                  rank_THORAX + treatment,
                  data = partridge)
anova(rank_ancova)
```

Analysis of Variance Table

```
Response: rank_LONGEV
      Df Sum Sq Mean Sq F value    Pr(>F)
rank_THORAX    1  63622    63622 129.511 < 2.2e-16 ***
treatment      4  40447    10112  20.584 6.536e-13 ***
Residuals    119  58458      491
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# Compare p-values with parametric ANCOVA
cat("P-value for treatment effect (parametric):",
    round(anova(ancova_model)[2, "Pr(>F)"], 4), "\n")
```

P-value for treatment effect (parametric): 0

```
cat("P-value for treatment effect (rank-based):",
    round(anova(rank_ancova)[2, "Pr(>F)"], 4))
```

P-value for treatment effect (rank-based): 0

```
# Permutation test example using lmPerm package
# library(lmPerm)
# perm_ancova <- lmp(LONGEV ~
#                   THORAX + treatment,
#                   data = partridge,
#                   perm = "Prob")
# summary(perm_ancova)
```

Note: The permutation test is commented out as it requires the `lmPerm` package, which may not be installed. The rank-based approach is shown as a simple alternative.

Writing Up ANCOVA Results

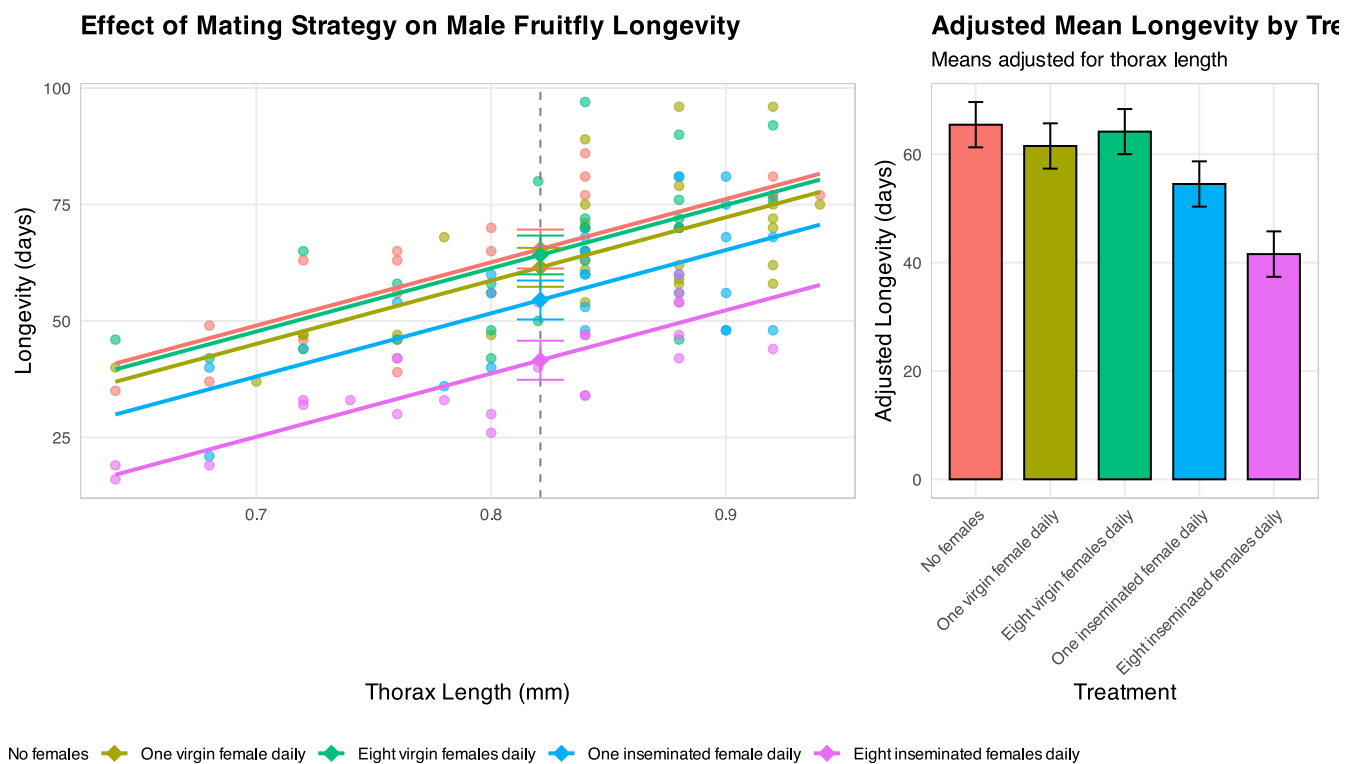
Scientific Writing Example

Here's how you might write up ANCOVA results for publication:

“We analyzed the effects of mating strategy on male fruitfly longevity using analysis of covariance (ANCOVA), with thorax length as a covariate. Before conducting the main analysis, we tested the homogeneity of slopes assumption and found no significant interaction between treatment and thorax length ($F_{4,115} = 1.56$, $P = 0.19$), indicating that the effect of body size on longevity was consistent across treatments.

The ANCOVA revealed significant effects of both treatment ($F_{4,119} = 27.97$, $P < 0.001$) and thorax length ($F_{1,119} = 145.44$, $P < 0.001$) on longevity. Thorax length was positively associated with longevity ($b = 1.19$), with larger males living longer. After adjusting for body size, males with no female partners lived significantly longer (adjusted mean $\pm$ SE: $1.81 \pm 0.02 \log_{10}$ days) than males in any other treatment group. Males provided with a single virgin female daily (1.77 ± 0.02) or a single inseminated female daily (1.79 ± 0.02) showed intermediate longevity, while males with eight females per day showed the lowest longevity (1.72 ± 0.02 for inseminated females; 1.59 ± 0.02 for virgin females). Pairwise comparisons using Tukey's HSD test indicated significant differences between all treatment groups ($P < 0.05$) except between the two treatments with a single female per day ($P = 0.42$).”

Publication Quality Figure



Summary

Key Principles

1. Purpose:

- ANCOVA combines regression and ANOVA approaches
- Increases power by accounting for continuous covariates
- Allows comparison of adjusted means

2. **The Analysis**

- Always test for homogeneity of slopes first!
- If slopes are homogeneous, proceed with standard ANCOVA
- If slopes are heterogeneous, use alternatives (Johnson-Neyman procedure)

3. **Interpretation**

- Focus on adjusted means (at mean covariate value)
- Consider both statistical and biological significance
- Visualize results clearly with appropriate graphs

Assumptions

1. Independence of observations
2. Normal distribution of residuals
3. Homogeneity of variances
4. Linearity of relationships within groups
5. Homogeneity of regression slopes